FPGA Programming – VHDL Part

VHDL by Examples – FSM

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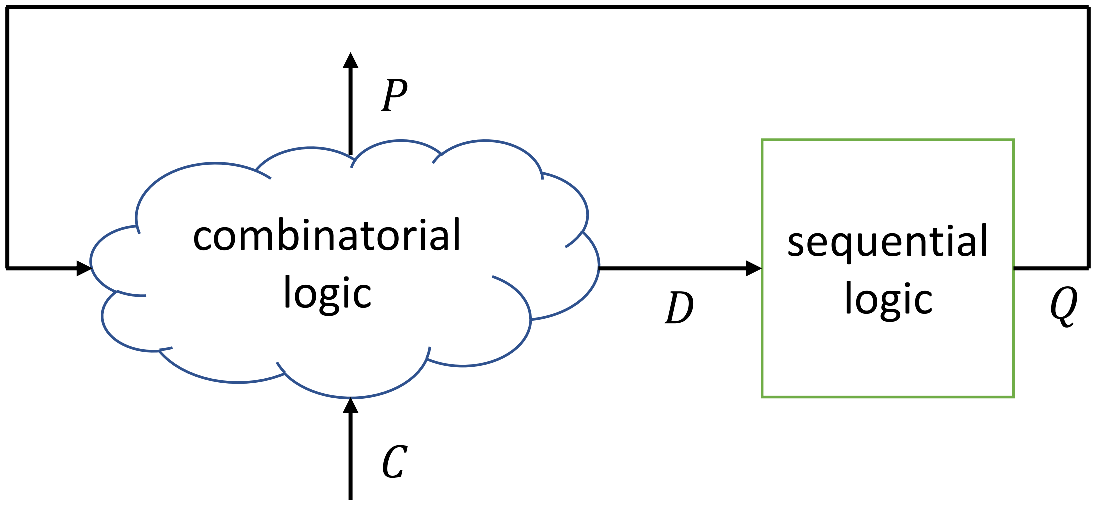
State machine

- A general schema for functions that control sequential logic take into account external signals C and produce control signals P
  - Mealey state machine: P is function of C and Q
  - Moore state machine: P is only function of Q
- Moore outputs are synchronous



VHDL state machine

- There are several ways of encoding a state machine
  - One process (clocked process with a case statement)
  - Two processes (clocked process for changing state, combinatorial process for setting outputs)
  - Three processes (clocked process for changing state, combinatorial process for next state choice, combinatorial process for setting outputs)
- Other combinations of processes are also allowed
- Just a coding style

VHDL state machine

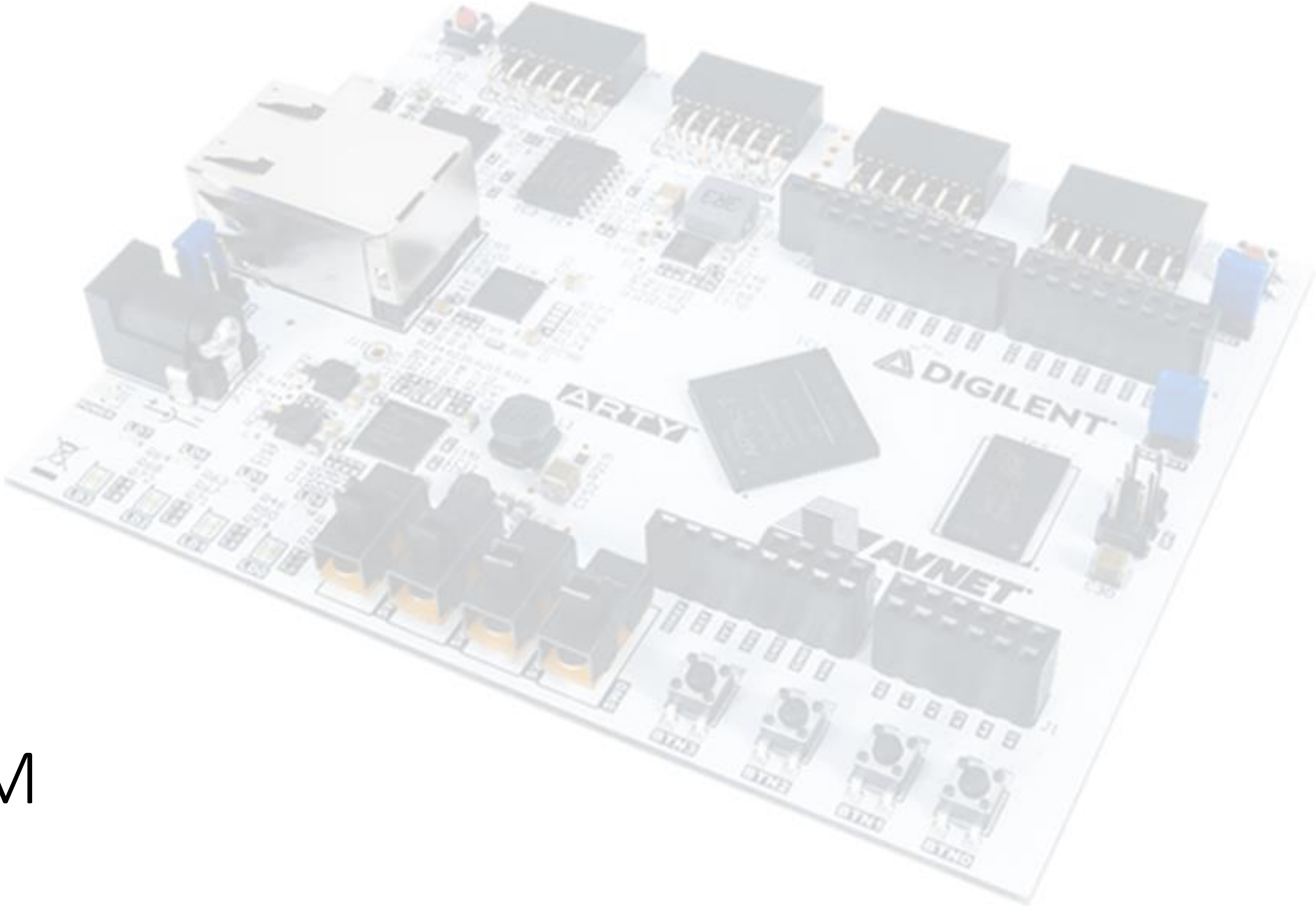
- One process state machine

```
process(Clk) is
begin
  if rising_edge(Clk) then
    if Rst = '1' then
      State <= S0;
      Dout  <= Value0;
    else
      case State is
        when S0 =>   if Condition0 then
                        State <= S1;
                        Dout <= Value1;
                     end if;
        when S1 =>   if Condition1 then
                        State <= S0;
                        Dout <= Value0;
                     end if;
        when others =>  Dout <= Value0;
                        State <= S0;
      end case;
    end if;
  end if;
end process;
```

VHDL state machine

- Two processes state machine

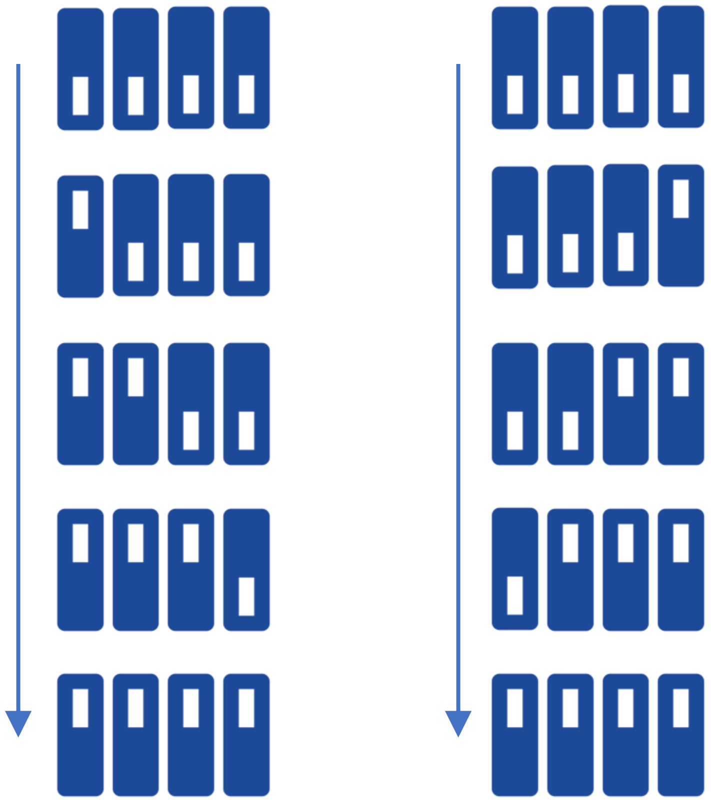
```
process(Clk) is
begin
  if rising_edge(Clk) then
     if Rst = '1' then
        State <= S0;
     else
        State <= NextState;
     end if;
  end if;
end process;

process(State, Condition0, Condition1) is
begin
   NextState <= State;

        case State is
          when S0 =>   Dout <= Value0;
                       if Condition0 then
                          NextState <= S1;
                       end if;
          when S1 =>   Dout <= Value1;
                       if Condition1 then
                          NextState <= S0;
                       end if;
          when others =>  Dout <= Value0;
                          NextState <= S0;

        end case;
      end if;
    end if;
  end process;
```



LAB FSM

FSM detecting sequences

- Design an FSM able to detect both the switching on of the four switches from left to right and from right to left
- Simulate
- Implement



EX FSM